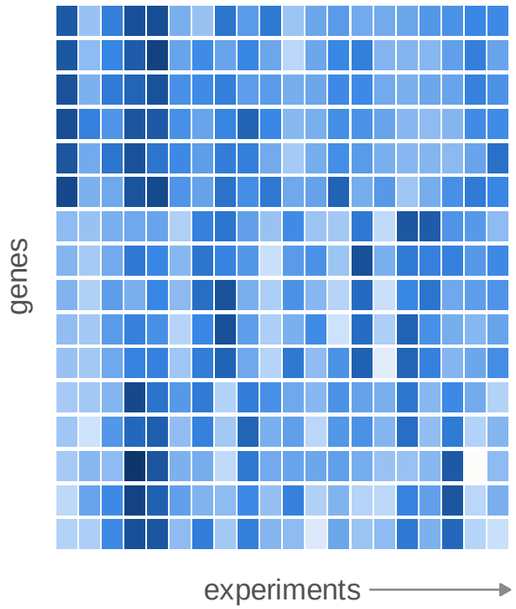
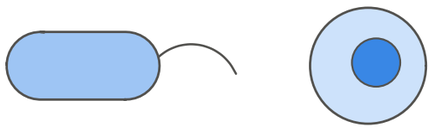
CLR (2007) is context-normalized attention; completing it, with no learning, recovers large regulatory programs

The gain requires context normalization; depth trades pair-level resolution for module-level recovery.

1 Expression compendium





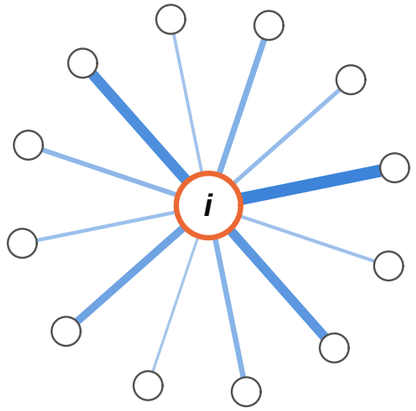
E. coli

M3D · EcoMAC microarrays

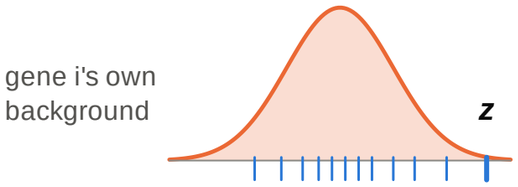
human

K562 Perturb-seq pseudobulk

2 CLR = context-normalized attention



edge width = mutual information

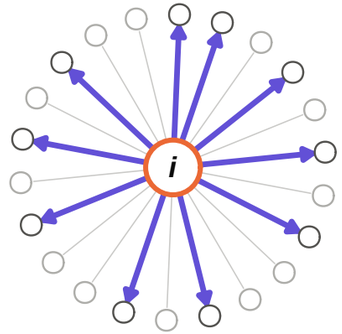


dependence ≈ query · key

context normalization ≈ scaled logits

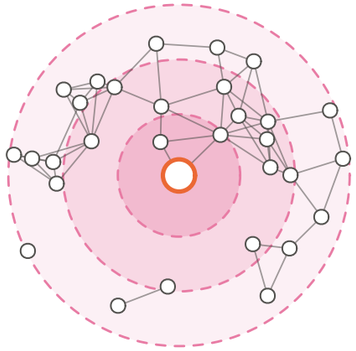
CLR (2007): attention before transformers

3 Completing the attention



softmax

weight on ~10 partners



multi-step walk

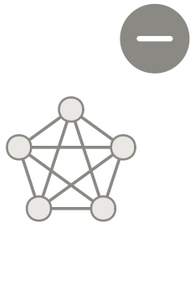
depth t (log)

1 3 9 27

no learned parameters

4 What it recovers





large regulatory programs ↑

E. coli regulons

tight complexes: CLR better

K562 CORUM: attention 3.51 · CLR 6.39

E. coli (M3D), strong-evidence regulons

AUPR / base rate (1 = random); t = 9

